

NATIONAL UNIVERSITY OF SINGAPORE

CS4268 – QUANTUM COMPUTING

(Semester 2: AY2022/23)

Time Allowed : 2 Hours

INSTRUCTIONS TO STUDENTS

1. Please write your Student Number only. Do not write your name.
2. This assessment paper contains **FOUR** questions for a total of **40** marks, and comprises **TWO** printed pages (excluding this page).
3. Students are required to answer **ALL** questions.
4. Students should write the answers for each question on a new page.
5. This is an OPEN BOOK assessment.
6. Permitted devices: electronic calculators

The following information is useful for all questions.

1. The Pauli matrices are given by

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

2. Let N be an integer greater than 1, and define the following sets:

$$\mathbb{Z}_N = \{0, \dots, N-1\} \quad ; \quad \mathbb{Z}_N^* = \{a \in \mathbb{Z}_N \mid \gcd(a, N) = 1\}.$$

3. For natural number $n \geq 2$, the Quantum Fourier Transform matrix QFT_n is defined as follows:

$$QFT_n = \frac{1}{\sqrt{n}} \sum_{x=0}^{n-1} \sum_{y=0}^{n-1} \omega_n^{xy} |y\rangle \langle x| ,$$

where $\omega_n = e^{\frac{2\pi i}{n}}$.

4. For integers $n \geq 2$, $j \neq 0$: $\sum_{i=0}^{n-1} \omega_n^{ij} = 0$.
5. Controlled-Controlled-i (CCI) gate: $CCI|x\rangle|y\rangle|z\rangle = i^{x \wedge y}|x\rangle|y\rangle|z\rangle$.
6. Phase-Toffol (PT) gate: $PT|x\rangle|y\rangle = (-1)^{x \wedge y}|x\rangle|y\rangle$.

Question 1. (10 marks)

Suppose Alice, Bob, and Charlie each have one qubit of the following three qubit state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|000\rangle_{ABC} + |111\rangle_{ABC}).$$

Alice, Bob and Charlie are allowed to do operations on their respective qubits and additionally they can send classical bits to each other. Describe what Alice, Bob and Charlie can do to ensure that the joint state that Alice and Bob are left with is the following state:

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}).$$

Justify your answer showing why it produces the state $|\phi\rangle$.

Question 2. (4+3+3 marks)

Let U be the operation corresponding to the circuit in Fig. 1. Show how the following gates can be performed by a circuit composed only of U gates. You are required to present circuits for each case. You can use as many ancilla qubits as you like, but they must start and end in the $|0\rangle$ state. You can use the circuit that you have constructed in any of the parts as a blackbox in the subsequent parts.

- (a) Hadamard gate.
- (b) Controlled-Controlled-i gate.
- (c) phase-Toffoli gate.

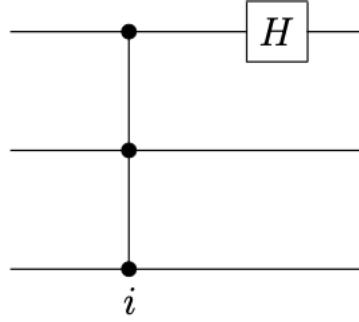


Figure 1

Question 3. (5+5 marks)

Let $f : \{0,1\}^n \rightarrow \{0,1\}$ be a periodic function such that $f(x) = f(x + r)$, for some unknown $1 < r < 2^n$. Define

$$|\hat{f}(l)\rangle = \frac{1}{\sqrt{r}} \sum_{x=0}^{r-1} \omega_r^{-lx} |f(x)\rangle.$$

1. Show that

$$|f(x)\rangle = \frac{1}{\sqrt{r}} \sum_{l=0}^{r-1} \omega_r^{lx} |\hat{f}(l)\rangle.$$

2. Let U_y be an operator such that $U_y |f(x)\rangle = |f(x+y)\rangle$. Show that $|\hat{f}(l)\rangle$ are eigenvectors of U_y . Find the corresponding eigenvalues.

Question 4. (10 marks)

Let $V_{a,b}$ be a unitary operator defined as follows (below $a, b, x, y, z \in \{0, 1\}$):

$$V_{a,b} |x\rangle |y\rangle |z\rangle = |x\rangle |y\rangle |z \oplus ((\overline{x \oplus a}) \wedge (\overline{y \oplus b}))\rangle.$$

Suppose you are given a black box implementing a unitary $V_{a,b}$ for some unknown a, b . Show that it is possible to find out a, b , using $V_{a,b}$ only once. You need to give an explicit circuit for solving the problem.

END OF PAPER